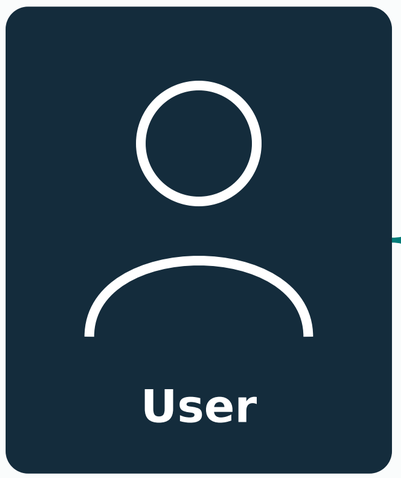
MailFlow AI: Data Design

Data that enters and leaves MailFlow through the user's interactions

TEAM

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1 Log in with Google

DATA IN

Google sign-in approval

DATA OUT / STORED

Google identity and OAuth tokens
→ User, ConnectedAccount

2 Open the main page and inbox

DATA IN

Gmail messages and labels (synced)

DATA OUT / STORED

Inbox list: subject, sender, priority, read/star state → Email, Label, EmailLabel

3 Read, star, or delete an email

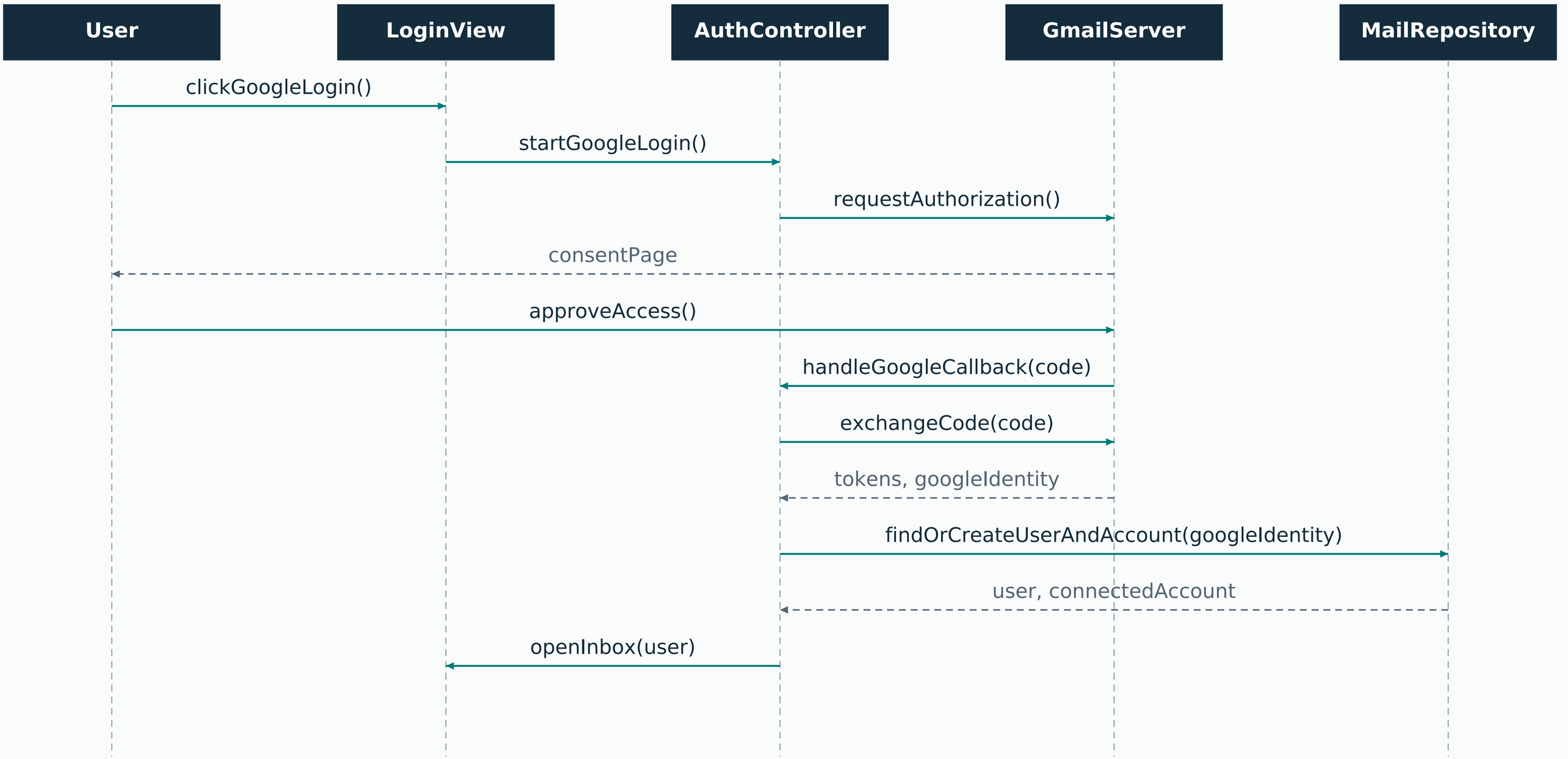
DATA IN

Selected email and action

DATA OUT / STORED

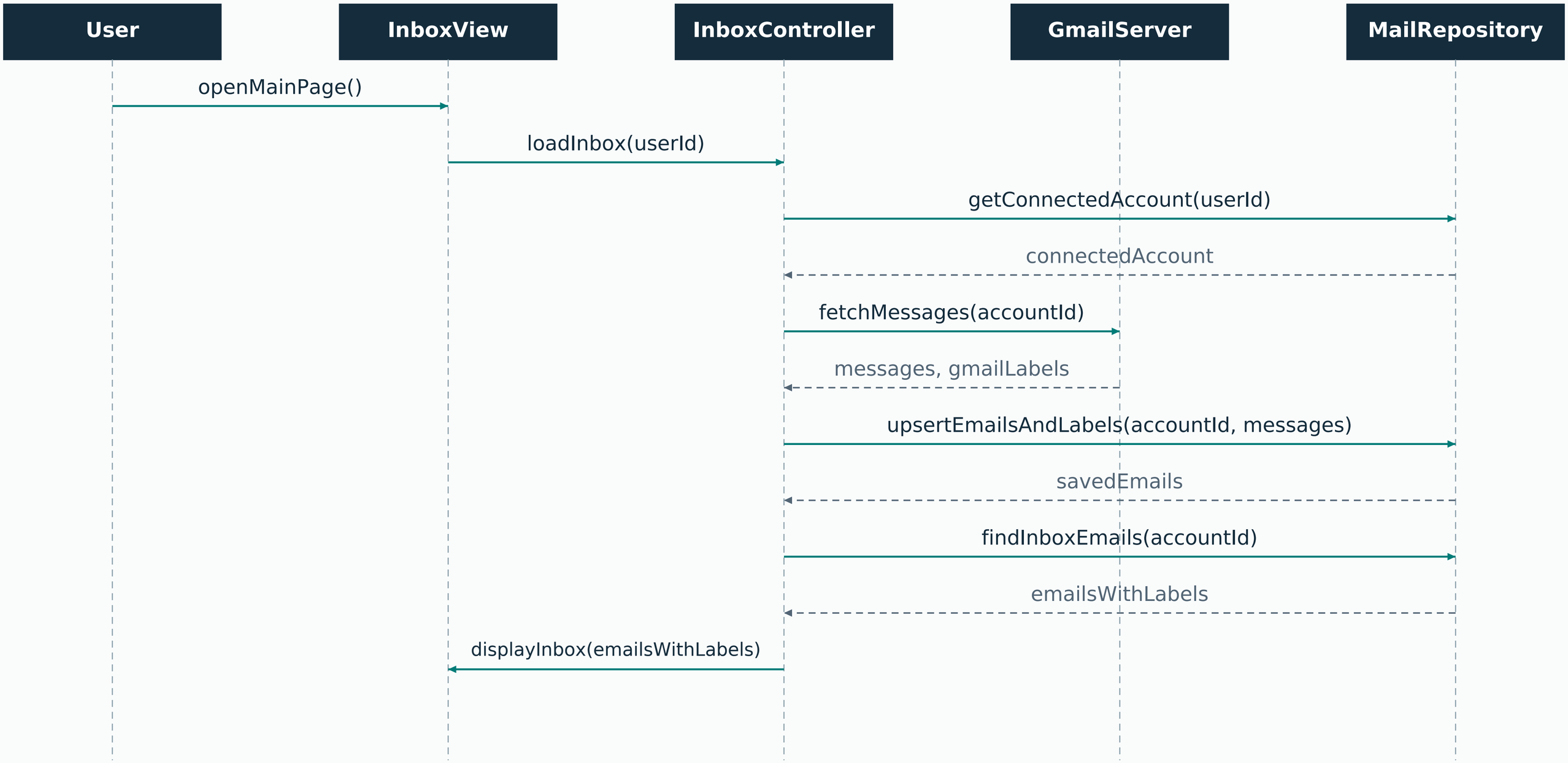
Updated read, starred, or trashed state
→ Email

1. User login



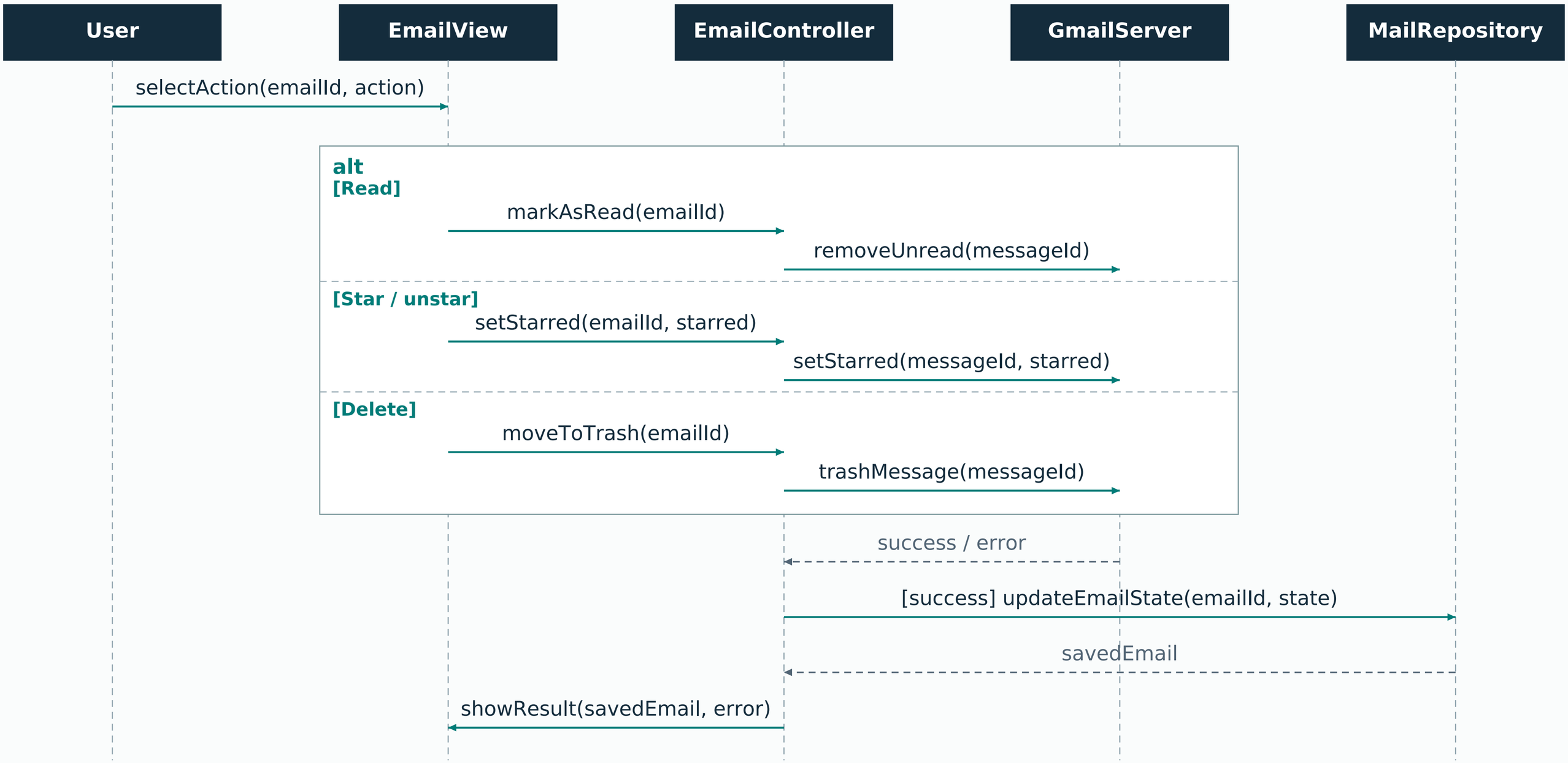
Google handles the password. AuthController establishes the application session after successful authorization.

2. Main page and inbox



The inbox shows subject, sender, priority, and read/star states. Repeated syncs update existing records using Gmail message IDs.

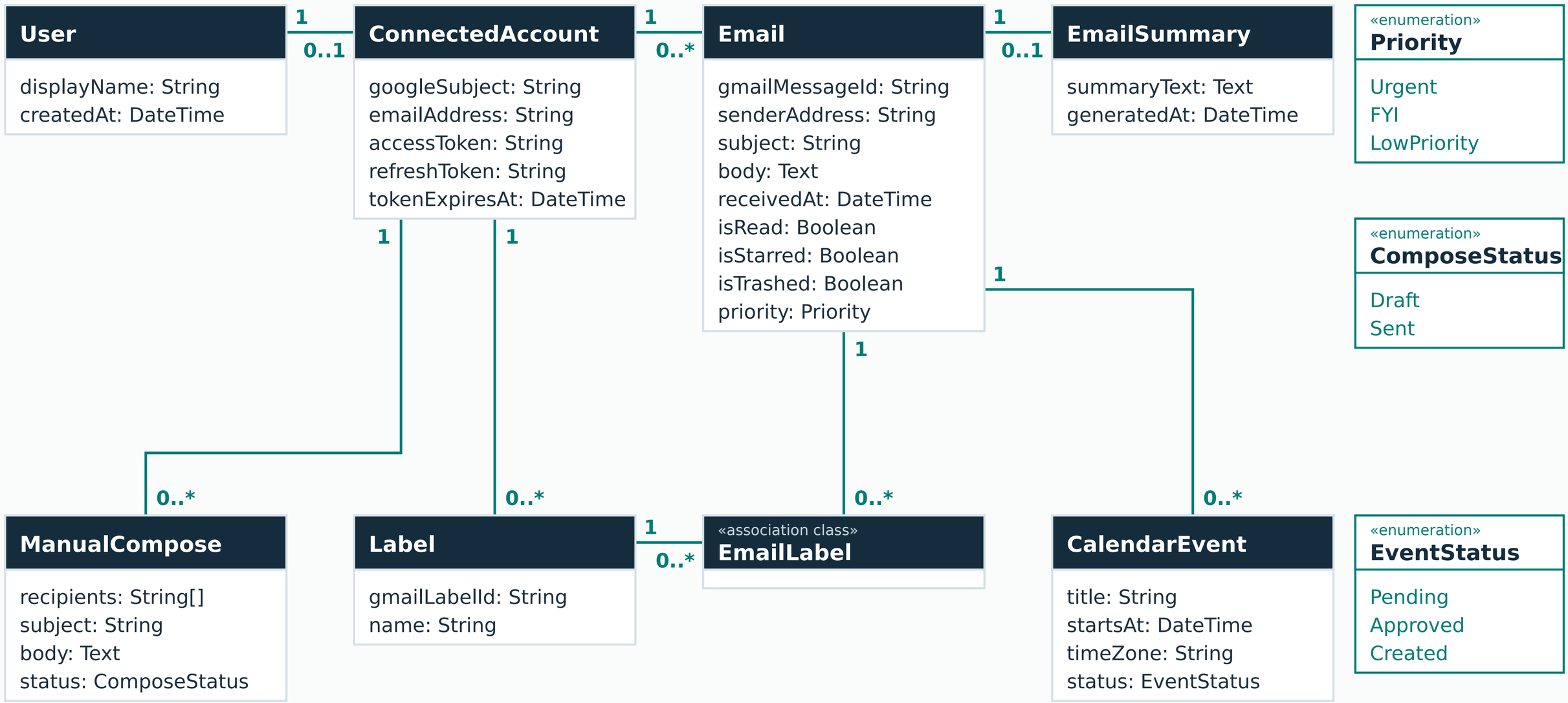
3. Read, star, or delete an email



The controller first resolves accountId and gmailMessageId from the selected email. On failure, it skips the state update and shows an error.

4. Class diagram (persisted data)

Data stored by MailFlow and mapped to the database with an ORM



Name descriptions

Meanings of class and attribute names

ConnectedAccount	The Google account a user links to MailFlow. MailFlow reads Gmail and Calendar through it.
googleSubject	Google's permanent, unique ID for the account, returned at sign-in.
accessToken / refreshToken	OAuth credentials for calling Gmail and Calendar; the refresh token renews an expired access token. Stored encrypted.
gmailMessageId, gmailLabelId	Gmail's own IDs. Re-syncing updates the existing record instead of creating a duplicate.
EmailLabel	Association class: one email can have many labels, and one label can cover many emails.
priority (Priority)	Category assigned to each email for the inbox and Daily Briefing: Urgent, FYI, or LowPriority.
isTrashed	The email was moved to Trash in Gmail; it is not permanently deleted.
EmailSummary	AI-generated summary of one email; generatedAt records when it was created.
ManualCompose	An email the user writes in MailFlow; kept as a Draft until it is Sent.
CalendarEvent	An event detected in an email, held as Pending until the user approves it and it is Created in Google Calendar.

GmailServer, CalendarServer, the controllers, and MailRepository are not stored as data.